

Assessing the Network with Common Security Tools (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 01

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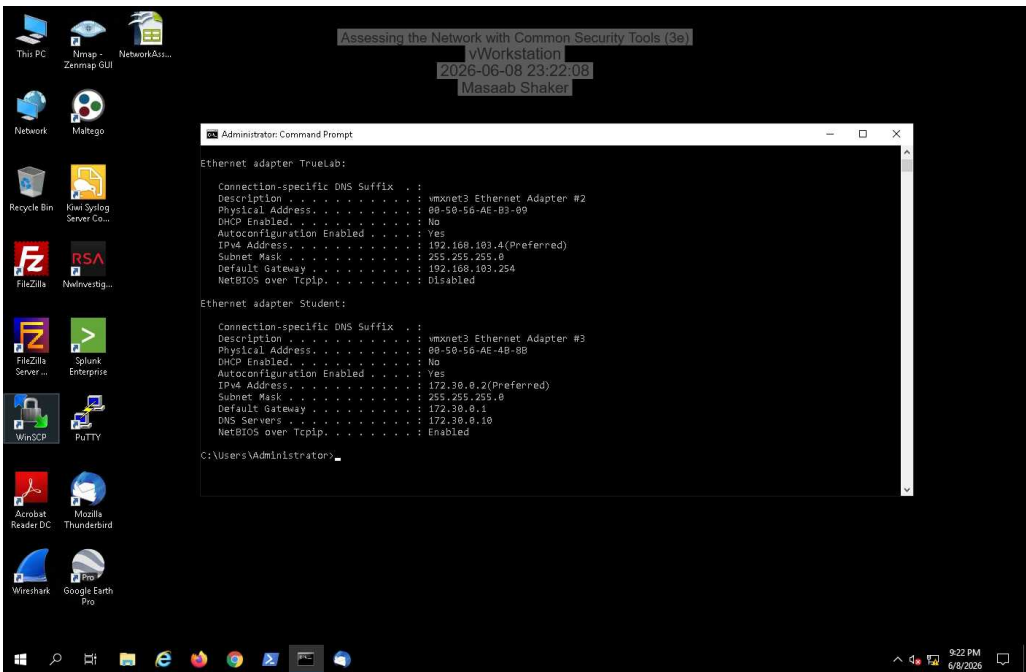
Time on Task:	Progress:
4 hours, 7 minutes	100%

Report Generated: Tuesday, June 9, 2026 at 3:42 AM

Section 1: Hands-On Demonstration

Part 1: Explore the Local Area Network

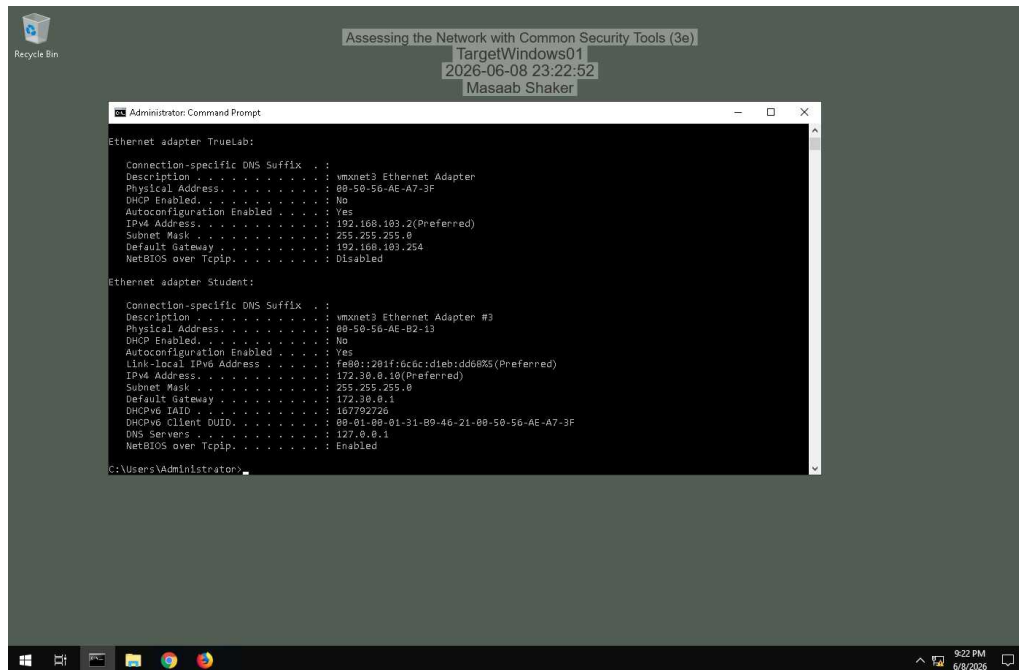
4. Make a screen capture showing the **ipconfig** results for the Student adapter on the vWorkstation.



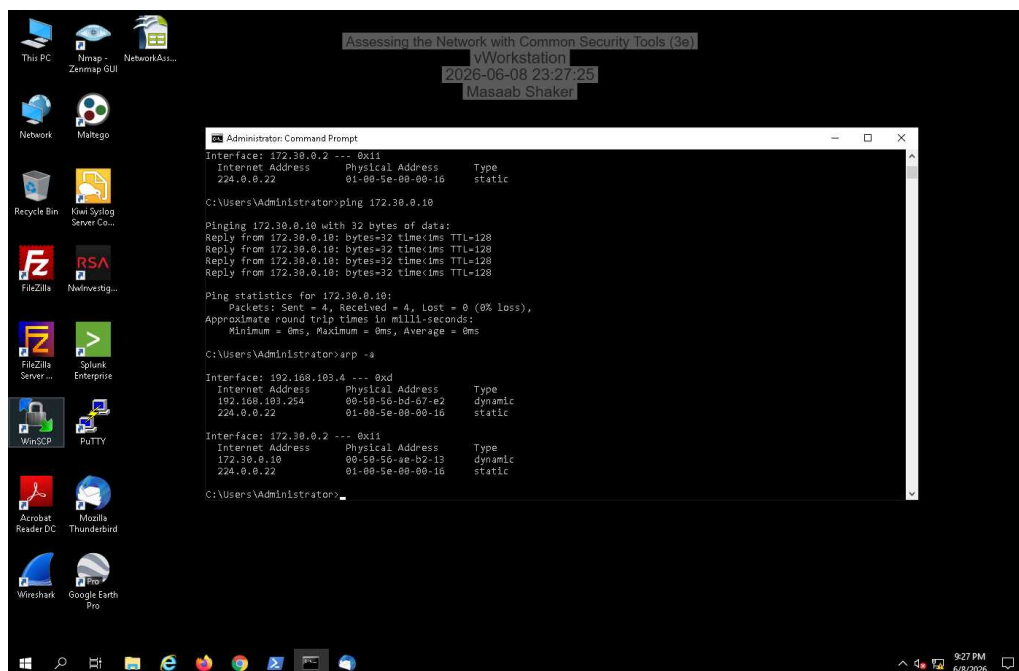
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7. Make a screen capture showing the **ipconfig** results for the Student adapter on **TargetWindows01**.



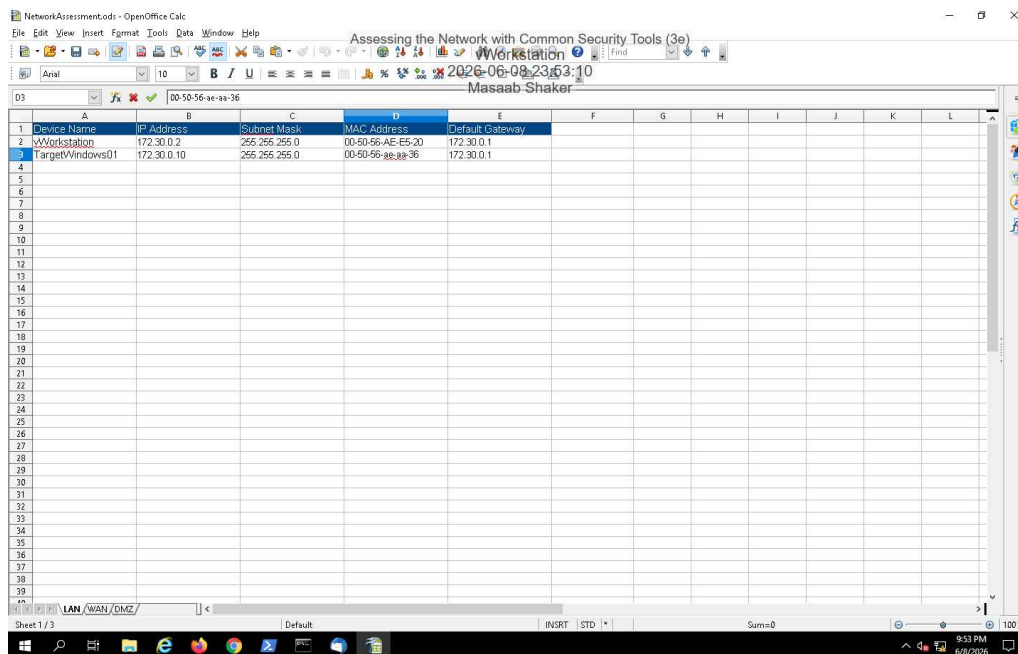
15. Make a screen capture showing the updated ARP cache on the vWorkstation.



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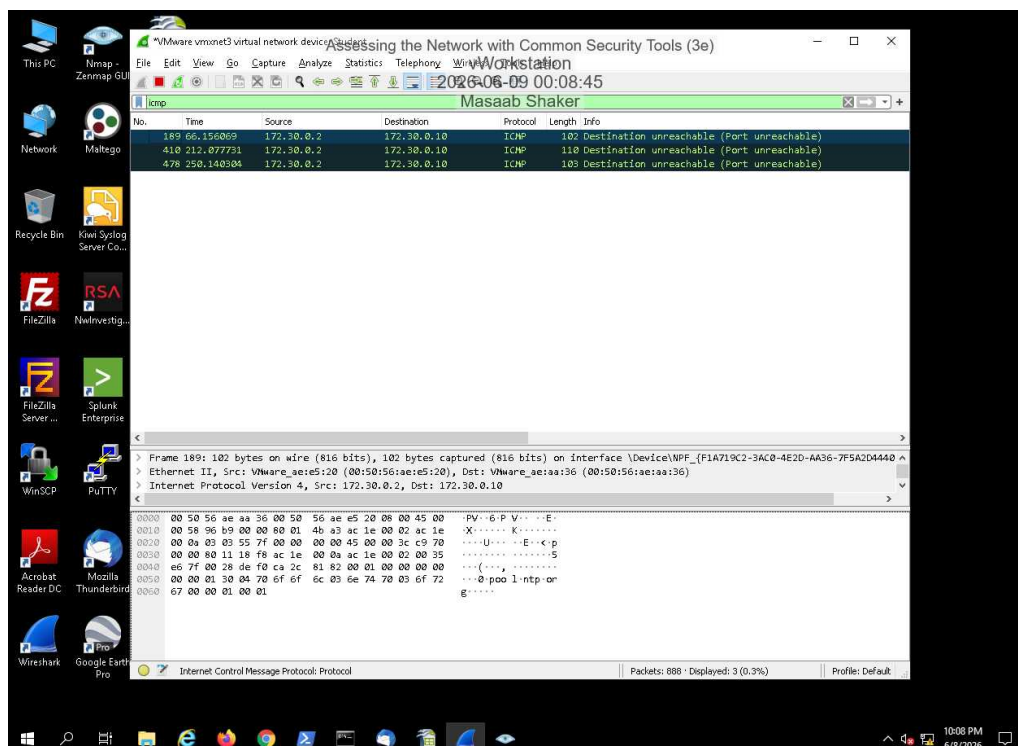
19. Make a screen capture showing the **completed LAN tab** of the Network Assessment spreadsheet.



Device Name	IP Address	Subnet Mask	MAC Address	Default Gateway
Workstation	172.30.0.2	255.255.255.0	00-50-56-AE-E5-20	172.30.0.1
TargetWindows01	172.30.0.10	255.255.255.0	00-50-56-9E-3A-36	172.30.0.1

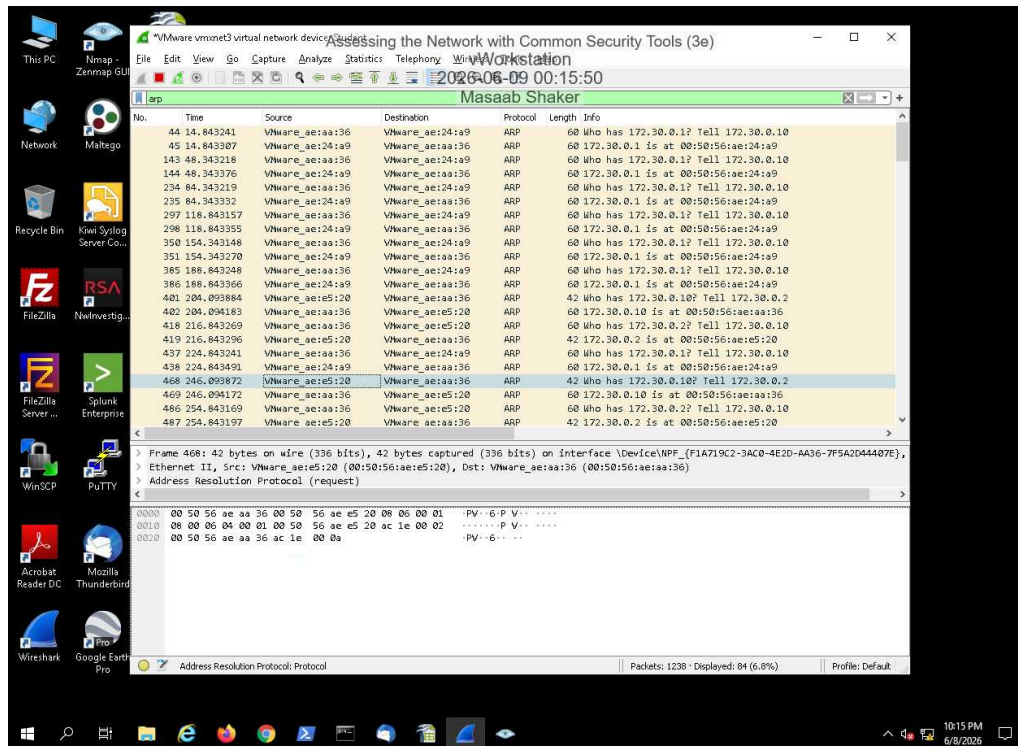
Part 2: Analyze Network Traffic

9. Make a screen capture showing the **ICMP filtered results** in Wireshark.



No.	Time	Source	Destination	Protocol	Length	Info
189	66.156869	172.30.0.2	172.30.0.10	ICMP	102	Destination unreachable (Port unreachable)
410	212.877731	172.30.0.2	172.30.0.10	ICMP	110	Destination unreachable (Port unreachable)
478	250.140384	172.30.0.2	172.30.0.10	ICMP	103	Destination unreachable (Port unreachable)

12. Make a screen capture showing the ARP filtered results in Wireshark.



18. **Compare** the Regular scan results for ICMP and ARP traffic with the results from the Ping scan.

Zenmap Regular scan: Starting Nmap 7.70 (<https://nmap.org>) at 2026-06-08 22:18 Pacific Daylight Time

Nmap scan report for pfsense.securelabsondemand.com (172.30.0.1)

Host is up (0.00s latency).

Not shown: 997 filtered ports

PORT STATE SERVICE

22/tcp open ssh

53/tcp open domain

80/tcp open http

MAC Address: 00:50:56:AE:24:A9 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.11 seconds

ZENMAP PING: Starting Nmap 7.70 (<https://nmap.org>) at 2026-06-08 22:22 Pacific Daylight Time

Nmap scan report for pfsense.securelabsondemand.com (172.30.0.1)

Host is up (0.00s latency).

MAC Address: 00:50:56:AE:24:A9 (VMware)

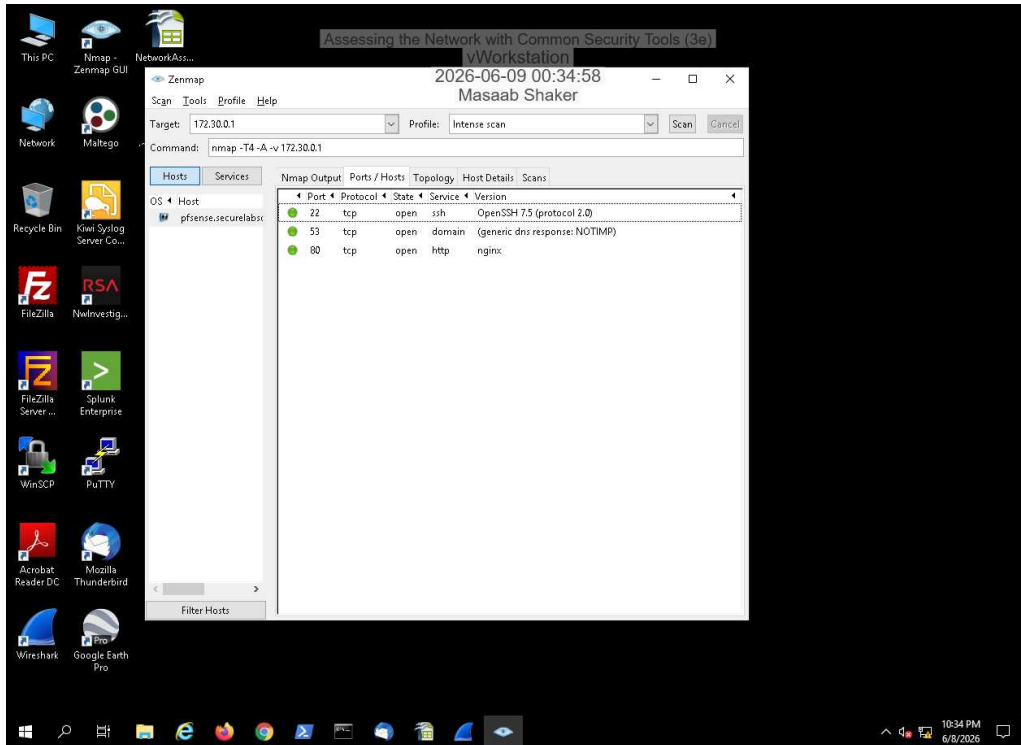
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

-----WIRESHARK:When using the regular scan it showed more open ports and more traffic was provided, but when I used ping scan it showed less traffic. the results were the same for both when I filtered using ICMP.

24. **Compare** the Intense scan results with the results from the Ping scan.

When I did the intense scan there were more results for the ICMP and same goes for ARP. Ping did not have that much traffic compared to intense scan.

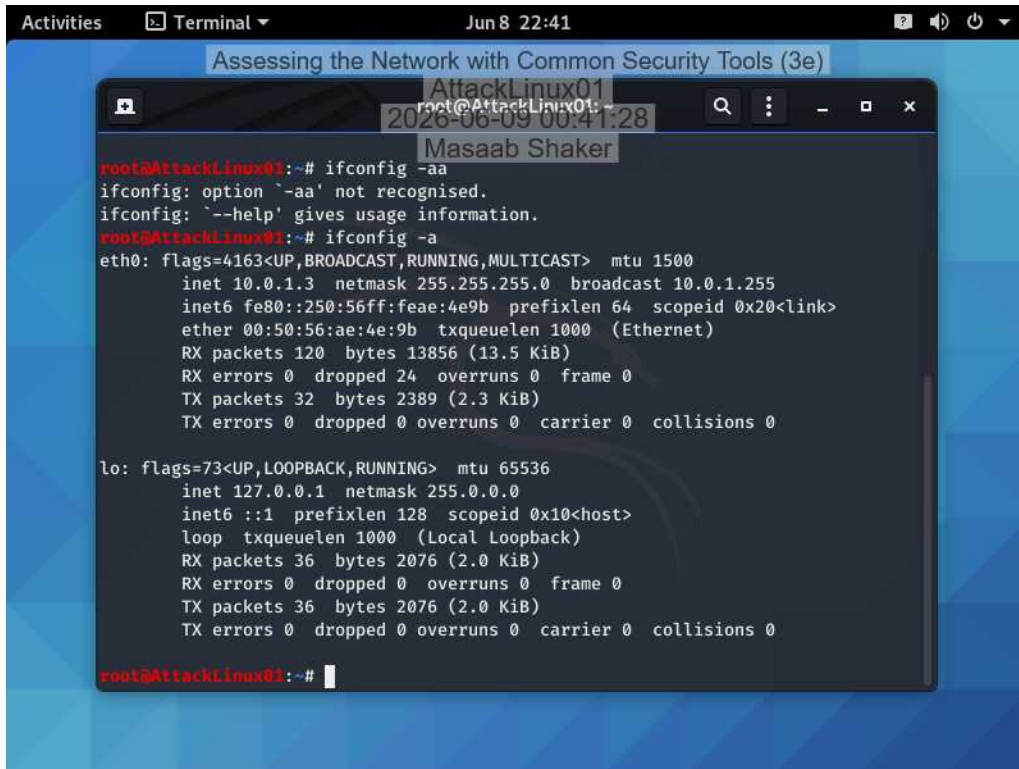
28. Make a screen capture showing the contents of the Ports/Hosts tab.



Section 2: Applied Learning

Part 1: Explore the Wide Area Network

6. Make a screen capture showing the `ifconfig` results on `AttackLinux01`.



The screenshot shows a terminal window titled "AttackLinux01" with the user "Masaab Shaker". The terminal displays the output of the `ifconfig -a` command. The output shows details for the `eth0` and `lo` interfaces. The `eth0` interface is configured with IP `10.0.1.3` and netmask `255.255.255.0`. The `lo` interface is the loopback interface with IP `127.0.0.1` and netmask `255.0.0.0`.

```
root@AttackLinux01:~# ifconfig -aa
ifconfig: option '-aa' not recognised.
ifconfig: '--help' gives usage information.
root@AttackLinux01:~# ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.1.3  netmask 255.255.255.0  broadcast 10.0.1.255
    inet6 fe80::250:56ff:feae:4e9b  prefixlen 64  scopeid 0x20<link>
    ether 00:50:56:ae:4e:9b  txqueuelen 1000  (Ethernet)
    RX packets 120  bytes 13856 (13.5 KiB)
    RX errors 0  dropped 24  overruns 0  frame 0
    TX packets 32  bytes 2389 (2.3 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 36  bytes 2076 (2.0 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 36  bytes 2076 (2.0 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

root@AttackLinux01:~#
```

12. Make a screen capture showing the ipconfig results on RemoteWindows01.

```
Administrator: Command Prompt
Assessing the Network with Common Security Tools (3e)
RemoteWindows01
2026-06-09 00:42:33
Masaab Shaker

Connection-specific DNS Suffix . : 
Description . . . . . : vmxnet3 Ethernet Adapter #3
Physical Address. . . . . : 00-50-56-AE-4C-84
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::f9a1:1060:22d7:1567%11(Preferred)
IPv4 Address. . . . . : 10.0.1.2(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.0.1.1
DHCPv6 IAID . . . . . : 469782614
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-B0-52-0C-00-50-56-AE-4C-84
DNS Servers . . . . . : fec0:0:0:ffff::1%1
                       : fec0:0:0:ffff::2%1
                       : fec0:0:0:ffff::3%1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter TrueLab:

Connection-specific DNS Suffix . : 
Description . . . . . : vmxnet3 Ethernet Adapter #2
Physical Address. . . . . : 00-50-56-AE-09-52
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
IPv4 Address. . . . . : 192.168.00.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.00.254
NetBIOS over Tcpip. . . . . : Disabled

C:\Users\Administrator>
```

18. Make a screen capture showing the updated ARP cache on RemoteWindows01.

```
Administrator: Command Prompt
Assessing the Network with Common Security Tools (3e)
RemoteWindows01
2026-06-09 00:49:23
Masaab Shaker

eth_addr Specifies a physical address.
If addr If present, this specifies the Internet address of the interface whose address translation table should be used. If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 ... Adds a static entry.
> arp -a ... Displays the arp table.

C:\Users\Administrator>arp -a

Interface: 10.0.1.2 --- 0xb
Internet Address Physical Address Type
224.0.0.22 01-00-5e-00-00-16 static

Interface: 192.168.00.1 --- 0xe
Internet Address Physical Address Type
192.168.00.254 00-50-56-ab-30-2a dynamic
224.0.0.22 01-00-5e-00-00-16 static

C:\Users\Administrator>ping 202.20.1.1

Pinging 202.20.1.1 with 32 bytes of data:
Reply from 202.20.1.1: bytes=32 time=63ms TTL=63
Reply from 202.20.1.1: bytes=32 time=63ms TTL=63
Reply from 202.20.1.1: bytes=32 time=63ms TTL=63
Reply from 202.20.1.1: bytes=32 time=63ms TTL=63

Ping statistics for 202.20.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\Users\Administrator>

C:\Users\Administrator>

C:\Users\Administrator>arp -a

Interface: 10.0.1.2 --- 0xb
Internet Address Physical Address Type
10.0.1.1 00-50-56-ab-1e-95 dynamic
10.0.1.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static

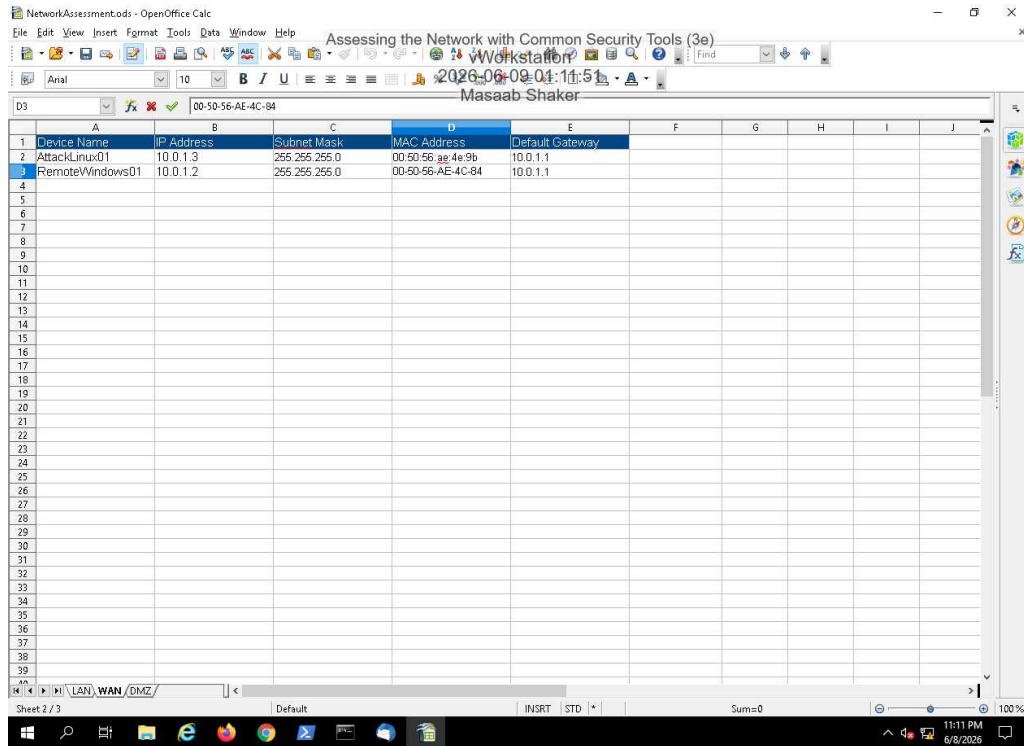
Interface: 192.168.00.1 --- 0xe
Internet Address Physical Address Type
192.168.00.254 00-50-56-ab-30-2a dynamic
224.0.0.22 01-00-5e-00-00-16 static

C:\Users\Administrator>
```


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22. Make a screen capture showing the **completed WAN tab** of the Network Assessment spreadsheet.

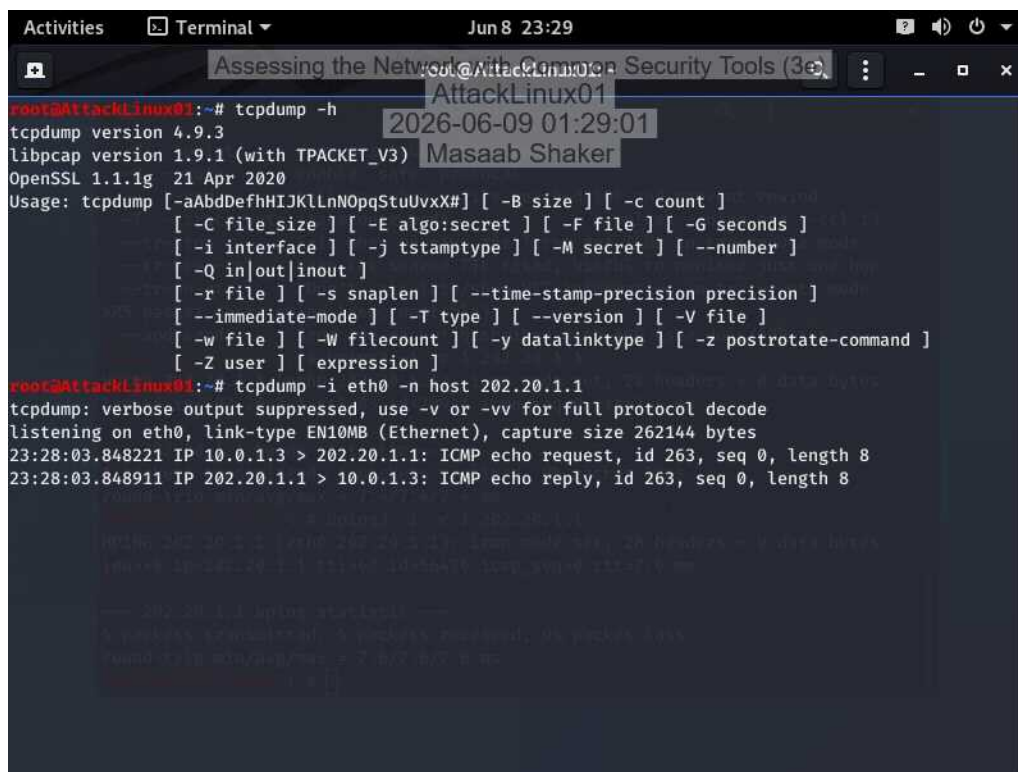


The screenshot shows an OpenOffice Calc spreadsheet titled "Assessing the Network with Common Security Tools (3e)". The spreadsheet is open to the "WAN" tab. The data is as follows:

Device Name	IP Address	Subnet Mask	MAC Address	Default Gateway
AttackLinux01	10.0.1.3	255.255.255.0	00:50:56:9a:4e:9b	10.0.1.1
RemoteWindows01	10.0.1.2	255.255.255.0	00:50:56:AE-4C-84	10.0.1.1

Part 2: Analyze Network Traffic

9. Make a screen capture showing **tcpdump** echo back the captured packets.

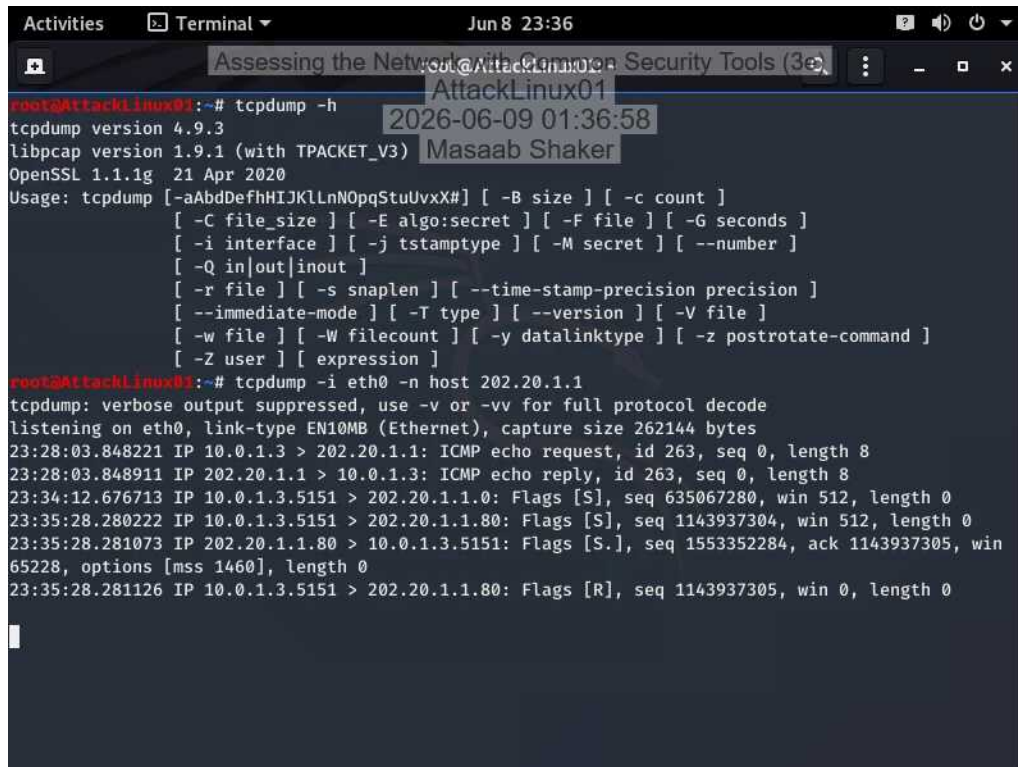


The screenshot shows a terminal window titled "Terminal" with the date and time "Jun 8 23:29". The terminal output is as follows:

```
root@AttackLinux01:~# tcpdump -h
tcpdump version 4.9.3
libpcap version 1.9.1 (with TPACKET_V3)
OpenSSL 1.1.1g  21 Apr 2020
Usage: tcpdump [-aAbDefhHIJKLlNOpqStuUvXx#] [-B size] [-c count] [-E algo:secret] [-F file] [-G seconds] [-I interface] [-j tstamptype] [-M secret] [--number] [-Q in|out|inout] [-r file] [-s snaplen] [--time-stamp-precision precision] [--immediate-mode] [-T type] [--version] [-V file] [-w file] [-W filecount] [-y datalinktype] [-z postrotate-command] [-Z user] [expression]

root@AttackLinux01:~# tcpdump -i eth0 -n host 202.20.1.1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
23:28:03.848221 IP 10.0.1.3 > 202.20.1.1: ICMP echo request, id 263, seq 0, length 8
23:28:03.848911 IP 202.20.1.1 > 10.0.1.3: ICMP echo reply, id 263, seq 0, length 8
```

12. Make a screen capture showing the attempted three-way handshake in tcpdump.

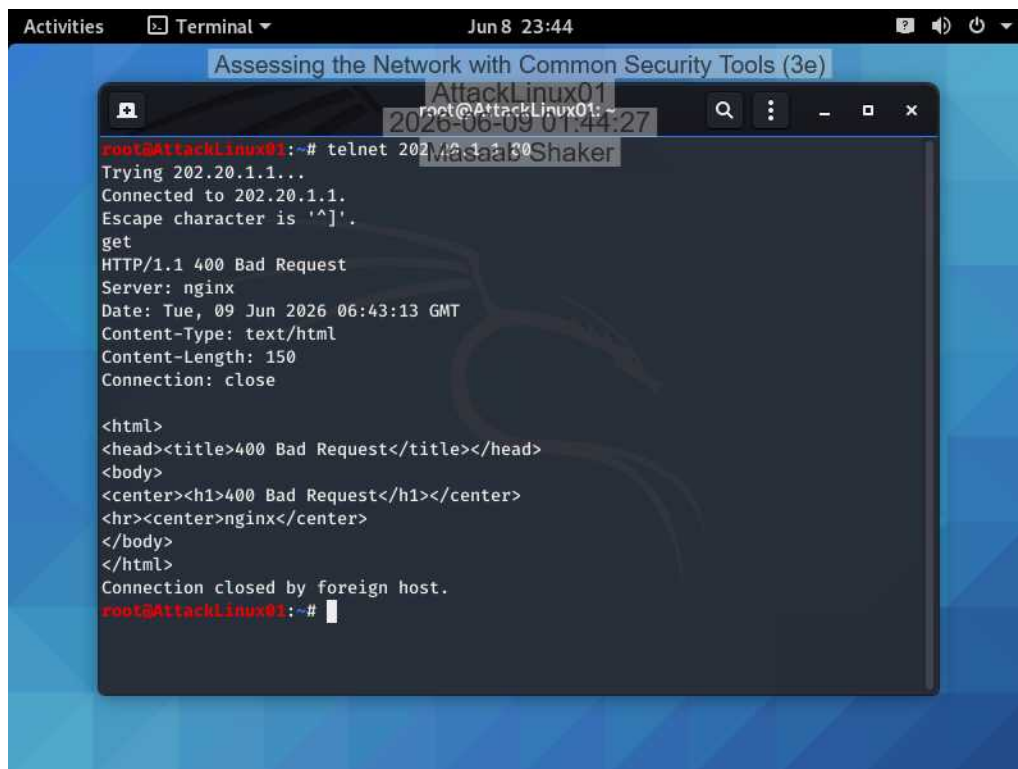


A terminal window titled "Assessing the Network with Common Security Tools (3e)" on a system named "AttackLinux01". The user runs the command `tcpdump -h`, displaying the help text for tcpdump version 4.9.3. Then, the user runs `tcpdump -i eth0 -n host 202.20.1.1`, which starts capturing traffic on the eth0 interface for host 202.20.1.1. The output shows several ICMP echo requests and replies, followed by three TCP flags [S] (SYN) from 10.0.1.3 to 202.20.1.1, indicating an attempted three-way handshake. The last line shows a TCP flags [R] (RST) from 10.0.1.3 to 202.20.1.1, indicating the connection attempt was rejected.

```
root@AttackLinux01:~# tcpdump -h
tcpdump version 4.9.3
libpcap version 1.9.1 (with TPACKET_V3)
OpenSSL 1.1.1g 21 Apr 2020
Usage: tcpdump [-aAbCdEfHhIjKlLnOpqStuUvxX#] [-B size] [-c count]
        [-C file_size] [-E algo:secret] [-F file] [-G seconds]
        [-i interface] [-j tstamptype] [-M secret] [--number]
        [-Q in|out|inout]
        [-r file] [-s snaplen] [--time-stamp-precision precision]
        [--immediate-mode] [-T type] [--version] [-V file]
        [-w file] [-W filecount] [-y datalinktype] [-z postrotate-command]
        [-Z user] [expression]

root@AttackLinux01:~# tcpdump -i eth0 -n host 202.20.1.1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
23:28:03.848221 IP 10.0.1.3 > 202.20.1.1: ICMP echo request, id 263, seq 0, length 8
23:28:03.848911 IP 202.20.1.1 > 10.0.1.3: ICMP echo reply, id 263, seq 0, length 8
23:34:12.676713 IP 10.0.1.3.5151 > 202.20.1.1.0: Flags [S], seq 635067280, win 512, length 0
23:35:28.280222 IP 10.0.1.3.5151 > 202.20.1.1.80: Flags [S], seq 1143937304, win 512, length 0
23:35:28.281073 IP 202.20.1.1.80 > 10.0.1.3.5151: Flags [S.], seq 1553352284, ack 1143937305, win
65228, options [mss 1460], length 0
23:35:28.281126 IP 10.0.1.3.5151 > 202.20.1.1.80: Flags [R], seq 1143937305, win 0, length 0
```

17. Make a screen capture showing the results of the get command.



A terminal window titled "Assessing the Network with Common Security Tools (3e)" on a system named "AttackLinux01". The user runs the command `telnet 202.20.1.1 80`, which connects to the host 202.20.1.1 on port 80. The user then enters the command `get`, which results in a "400 Bad Request" response from the nginx server. The response includes headers like "Server: nginx", "Date: Tue, 09 Jun 2026 06:43:13 GMT", "Content-Type: text/html", and "Content-Length: 150". The body of the response is an HTML document with a title "400 Bad Request" and a message "400 Bad Request" in the center. The connection is then closed by the foreign host.

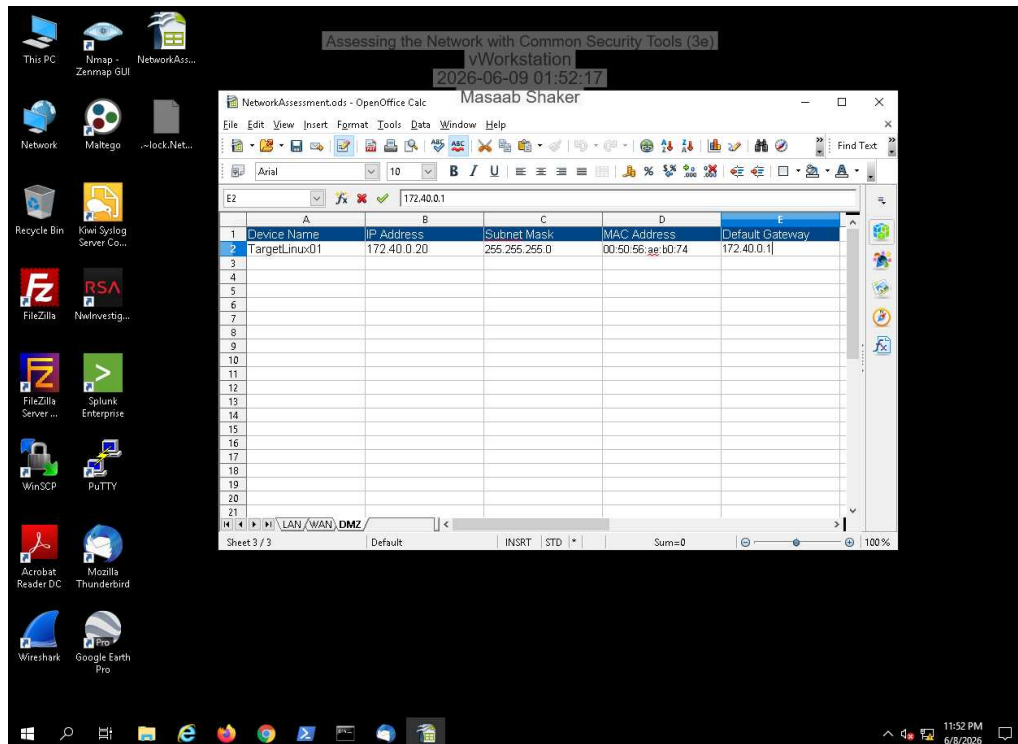
```
root@AttackLinux01:~# telnet 202.20.1.1 80
Trying 202.20.1.1...
Connected to 202.20.1.1.
Escape character is '^'.
get
HTTP/1.1 400 Bad Request
Server: nginx
Date: Tue, 09 Jun 2026 06:43:13 GMT
Content-Type: text/html
Content-Length: 150
Connection: close

<html>
<head><title>400 Bad Request</title></head>
<body>
<center><h1>400 Bad Request</h1></center>
<hr><center>nginx</center>
</body>
</html>
Connection closed by foreign host.
root@AttackLinux01:~#
```


Section 3: Challenge and Analysis

Part 1: Explore the DMZ

Make a screen capture showing the **completed DMZ tab** of the **NetworkAssessment** spreadsheet.



Part 2: Perform Reconnaissance on the Firewall

Briefly summarize and analyze your findings in a technical memo to your boss.

While using Zen Map I did a regular scan on the pfSense firewall on IP address of 172.30.0.1, and I was able to capture various packets using Wireshark. There was 4 ICMP packets that were captured when being sent to the firewall, 5+ ARP packets being sent, and 0 DNS packets sent to firewall. The ports that were open are 22(ssh), 53(domain), and 80(http).